

Question No.	Question Type	CILOs assessed	Weight	Marks	Marker signature	Student score
1	Drawing & Problems	CILO 4,5,6,7,9	60 %	30		
2	Multiple Choice	CILO1,2,7,8	40 %	20		
Total			100 %	50		

Moderation Review			
Comments			
Marks Reviewed by		Signature & Date	

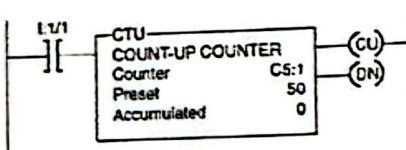
Q1 (20 Marks)

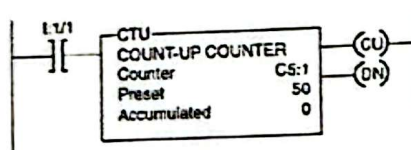
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Select the best correct answer, and write it in the table below.

1. The ladder logic in PLC consists of			
a- Logic gates	b- Relay contacts and coils	c- Functional blocks	d- Relays
2. An AND function implemented in ladder logic uses			
a- Normally-open contacts in series	b- Normally-closed contacts in series		
c- A single normally-closed contact	d- Normally-open contacts in parallel		



3. In PLCs rungs power flows from:			
a- The right rail to the left rail		b- The up rail to the left rail	
c- The down rail to the left rail		d- The left rail to the right rail	
4. Timers may have one or more _____ for their reference timing.			
a- Analog outputs	b- Analog inputs	c- Unlatch inputs	d- Time bases
5. In an up-counter, when the accumulated count exceeds the preset count without a reset, the accumulated count will:			
a- Hold the accumulated value.		b- Continue incrementing.	
c-Set itself to zero.		d- Start decrementing.	
6. For the following ladder diagram suppose your accumulated value is 24 and you lose AC line power to the controller. When power is restored to your controller, the counter accumulated value will be:			
			
		d-50.	



a- 24	b- 26	c- 0.	d- 50.
7. The length of time delay period in TOFF timer is calculated by:			
a- The product of the timer time value and the time base	b- The product of the preset value and the timer time.	c- The product of the preset value and the time base.	d- The product of the preset value and the accumulated value
8. The instruction list for the following ladder logic diagrams is:			
a- LD I0.0 EXP-W AC4, AC4	b- LD I0.0 EXP-W AC0, AC0.	c- LD I0.0 EXP-W AC4, AC0.	d- LD I0.0 EXP-W AC0, AC4.

9. The instruction list for the following ladder logic diagrams is:			
a- LD I0.0 CTUD C40, #5 LD I0.1 LD I0.2	b- LD I0.0 LD I0.1 CTUD C40, 5 LD I0.2	c- LD I0.0 LD I0.1 LD I0.2 CTUD C40, # 5	d- LD I0.0 LD I0.1 LD I0.2 CTU C40, # 5

10. A PLC down-counter (CTD) counter counts transitions										
a- true to true			b- false to true			c- false to zero		d- false to false		
Q. NO	1	2	3	4	5	6	7	8	9	10
Ans.										

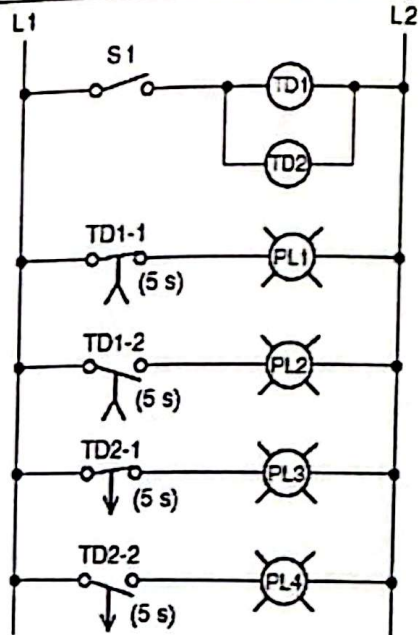


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Q2 (10 marks)

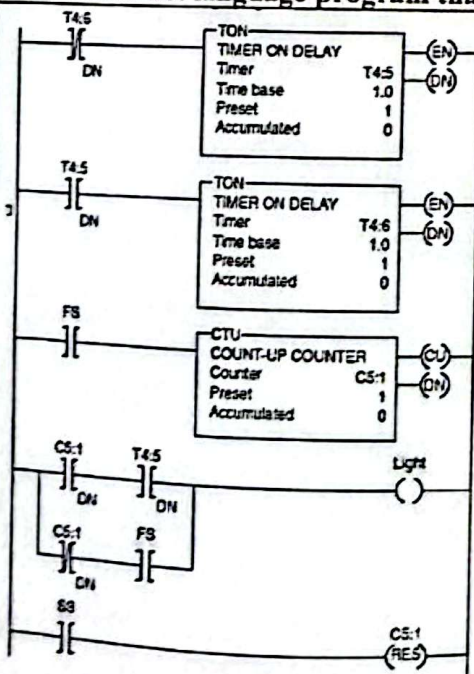
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With reference to the relay schematic diagram in the table using an Allen Bradley PLC unit, design a program using ladder diagram program language that will execute this hardwired control circuit correctly.



Q3 (10 marks)

With reference to the ladder diagram in the table using a Siemens S7-200 PLC unit, write the instruction list language program that will execute this ladder logic correctly.





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Q4 (10 marks)

Write a ladder diagram program using Allen Bradley PLC unit to convert Celsius temperature to Fahrenheit based on the following equation $F^{\circ} = C^{\circ} \times \frac{9}{5} + 32$

The program is designed to convert the recorded Celsius temperature in the data table to Fahrenheit values for display (O:13).

Mohammad Mahmoud